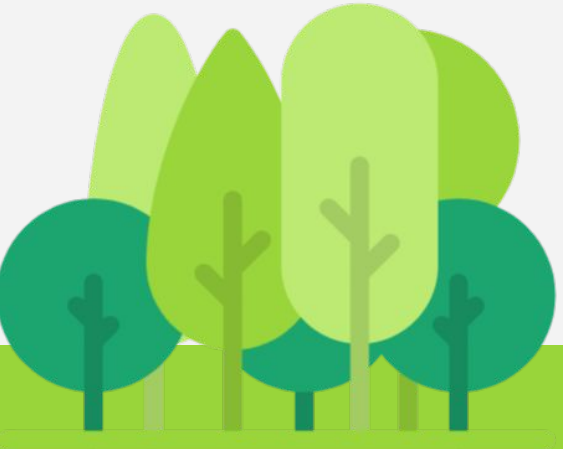
Random forest classifier



What is Random Forest?

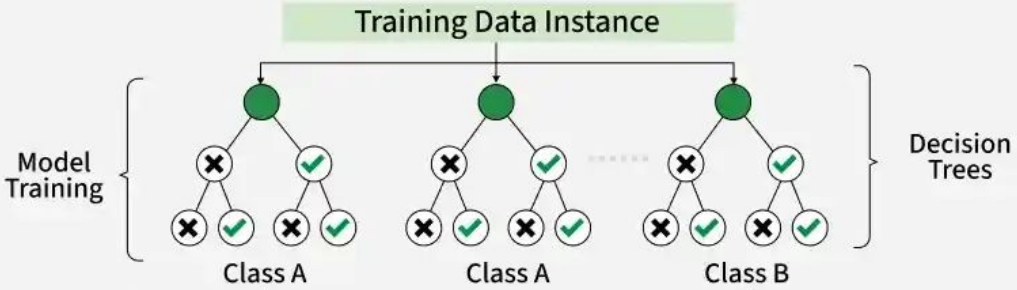
Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their outputs.

Purpose of Random Forest

- Improve prediction accuracy
- Reduce overfitting
- Handle large and complex datasets
- Provide robust and stable results

Random Forest Concept

1. Creating multiple trees
2. Making each tree different
3. Combining the results



https://www.geeksforgeeks.org/random-forest-classifier-using-scikit-learn/

How Random Forest Works

1. **Bootstrap Sampling**
   - Random samples are drawn from the dataset
2. **Feature Randomness**
   - At each split, a random subset of features is selected
3. **Build Multiple Trees**
   - Each tree is trained independently
4. **Voting**
   - Final output = majority vote

Example: Loan Approval Dataset

Tree 1 → Approved

Tree 2 → Rejected

Tree 3 → Approved

Tree 4 → Approved

Tree 5 → Rejected

Final prediction = Approved

Why Random Forest Works Well

- Reduces variance (compared to a single tree)
- Combines multiple weak learners into a strong learner
- Less sensitive to noise and outliers

Choosing Number of Trees

- Controlled by parameter: n_estimators
- Increase until performance stabilizes

Advantages

- High accuracy
- Reduces overfitting
- Works well with large datasets
- Handles missing values and noise
- Requires less parameter tuning

Disadvantages

- Slower than a single decision tree
- Requires more memory
- Less interpretable (black-box nature)

When to Use Random Forest

- When accuracy is more important than speed
- When data is complex and non-linear
- When you want a reliable baseline model

Applications

- Medical diagnosis
- Customer segmentation
- Credit risk analysis (Loan Approval)

Conclusion

- **Random Forest is a powerful ensemble method**
- **Combines multiple decision trees**
- **Improves accuracy and robustness**
- **Widely used in classification tasks, including loan approval predictions**

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Thank You

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